Abdelrahman Taeha

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EDUCATION

Georgia Institute of Technology — **Atlanta, GA**
Master of Science in Computer Science – Artificial Intelligence Concentration — *Aug. 2026 – May 2028 (Expected)*

Sacramento State University — **Sacramento, CA**
Bachelor of Science in Computer Science — *Aug. 2023 – Dec. 2025*

EXPERIENCE

Software Engineer — **May 2025 – Present**
DHCS — *Sacramento, CA*

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting mean time to detection (MTTD) for production incidents in half.
- Engineered backend integrations and validation workflows that surfaced system behavior anomalies early, accelerating root-cause analysis by 35% and streamlining stakeholder communication.
- Established internal dashboards and reporting applications to visualize agent-driven operational metrics, reducing incident resolution time by 30%.
- Implemented log-processing pipelines and monitoring integrations that improved real-time alerting accuracy by 20%.

AI Research Student — **August 2025 – Present**
Independent Research — *Remote*

- Evaluated medical LLM agent integrity across 840 paired-twin scenarios for the MedInsider FHIR-style benchmark under institutional pressure.
- Built evaluation and logging infrastructure — including deterministic action tracking and auditing workflows — for 9 distinct medical vision-language models across full-pipeline and baseline sub-panels.
- Trained a neutral baseline model and developed data collection pipelines for 2 major open-access clinical repositories used to analyze demographic bias in spinal radiology VLM outputs.
- Co-authored SpineFairBench, a counterfactual benchmark evaluating 7,996 paired spinal radiographs to audit demographic sensitivity in VLM-based report generation.

Software Engineering Intern (Machine Learning) — **May 2024 – Jul. 2025**
District Hut — *Sacramento, CA*

- Developed large-scale data pipelines processing 3 TB/week across distributed cloud infrastructure, improving throughput and system reliability in a high-volume production environment.
- Optimized data workflows and query performance, identifying and resolving bottlenecks that cut average pipeline latency and improved system stability under load.
- Revamped data integrity checks with SQL, preventing five critical production issues related to data corruption and automating 75% of data validation tasks with 100% test coverage.
- Collaborated with a team of 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week via autonomous performance metrics.

PROJECTS

AutoApply | *Python, Claude API, Flask, Web Scraping*

- Built a full-stack job-application toolkit that scrapes CalCareers and external boards, calls the Claude API to generate tailored resumes, cover letters, and SOQs per posting, and scores fit on a 0–100 scale.
- Engineered a human-in-the-loop review step and local web dashboard tracking every application through the pipeline — from discovery to submission.
- Automated document generation and tracking, cutting manual effort per application from ~30 minutes to under 5 minutes.

S.A.I.N.T. — Network Intrusion Detection System | *Python, PyTorch, Flask, Redis, NumPy*

- Teamed with 2 peers to develop an AI agent reasoning engine for threat detection, reaching 89% accuracy on NSL-KDD (125K+ samples).
- Engineered a decision-making layer using 41 traffic-flow features (protocol stats, TCP flag distributions, temporal patterns) to autonomously classify and flag malicious activity.
- Architected a low-latency inference agent using Flask + Redis + Plotly, processing 50–100 network connections/sec with an interactive agent-in-the-loop dashboard.

Al-Manar — Islamic Companion App (iOS) | *Flutter, Dart, Provider, Google Places API, Xcode Cloud*

- Led end-to-end development of a production iOS app built with Flutter and Dart, managing the full software lifecycle from UI/UX design through App Store distribution via Xcode Cloud CI/CD.
- Constructed a custom background audio framework extending BaseAudioHandler for seamless Quran recitation with system-level media controls, persistent playback state, and optimized memory handling.
- Integrated AI-driven location agents using the Google Places API and Haversine formula to deliver GPS-based prayer times and real-time Masjid discovery for a global user base in 9 languages.

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R, COBOL
Frameworks/Platforms: React.js, Node.js, Spring Boot, AWS, Docker, Git, GCP
Data & ML: TensorFlow, scikit-learn, pandas, NumPy, SAP (Data Integration), model evaluation
Security/DevOps: SIEM, GitHub Actions, CI/CD, monitoring pipelines
AI & Agents: Autonomous Agents, LLM Orchestration (LangChain/CrewAI), RAG, Prompt Engineering